

# HL7 interface

What the hospital's interface engine receives from OpenBedside, and what it may send.  
Written for the interface analyst. HL7 v2.6, MLLP framing, UTF-8.

## Connections

| Direction        | Transaction                                  | Message                                                       | Default                      | Acknowledgement                                               |
|------------------|----------------------------------------------|---------------------------------------------------------------|------------------------------|---------------------------------------------------------------|
| Gateway<br>→ EHR | IHE PCD-01,<br>Communicate<br>PCD Data       | ORU^R01^ORU_R01                                               | client to<br>127.0.0.1:6661  | application ACK<br>required ( MSH-15 NE ,<br>MSH-16 AL )      |
| EHR →<br>Gateway | IHE PCD-03,<br>Communicate<br>Infusion Order | RGV^015^RGV_015                                               | listens on<br>127.0.0.1:6662 | RRG^016 on the same<br>connection. <b>Simulator<br/>only.</b> |
| EHR →<br>Gateway | HL7 v2 ADT<br>(patient<br>location)          | ADT^A01 , A02 , A03 ,<br>A04 , A08 , A11 ,<br>A12 , A13 , A40 | listens on<br>127.0.0.1:6663 | ACK , AA for every<br>event, used or not                      |
| Gateway<br>→ EHR | IHE PCD-10,<br>Infusion Event                | ORU^R42^ORU_R01                                               | planned                      |                                                               |

Ports and hosts are set in the config file. Nothing listens on a network interface other than localhost until someone changes that on purpose.

## When PCD-01 is sent

Only for devices registered on the Devices page (unless [registry] unknown\_devices = "send" ), and then:

- whenever the engine derives an event on any channel of a device;
- whenever the device's patient, location or registration changes;
- otherwise every periodic\_seconds (default 60) per device.

Each message is a full snapshot of the device, not a delta.

## Containment in OBX-4

MDS.VMD.CHANNEL.METRIC . For module *m*:

| OBX-4   | Meaning                                           |
|---------|---------------------------------------------------|
| 1.0.0.0 | the device (MDS). OBX-18 identifies it, see below |

| OBX-4   | Meaning                                               |
|---------|-------------------------------------------------------|
| 1.m.0.0 | module <i>m</i> (VMD). OBX-18 carries the module id   |
| 1.m.1.0 | delivery channel: what the module is doing now        |
| 1.m.2.0 | primary source: what is hanging                       |
| 1.m.3.0 | secondary source, only when a piggyback is programmed |

Metrics inside a delivery channel: .1 status, .2 actual flow rate. Metrics inside a source: .1 drug name, .2 concentration, .3 programmed rate, .4 dose rate, .5 volume to be infused, .6 volume remaining, .7 volume delivered, .8 patient weight (only when the dose rate is weight-based).

## Device identity: OBX-18

identifier^vendor, plus ^EUI-64 value^EUI-64 when the device's registration has an EUI-64. The identifier is whatever the registration says the hospital scans and files by: the serial number, the hospital asset tag, or the EUI-64. Module ids follow it: BIO-004512-L for the left module of a pump registered by asset tag BIO-004512. This follows the form in published vendor PCD-01 specifications; check it against DEV TF-2 Appendix B before claiming conformance.

## Patient: PID and PV1

PCD-01 requires a PID segment (DEV TF-2 Rev 10.0, 3.1.4.1.1, PID R [1..1]), so there is always one. The Unknown form below is the one used in published vendor PCD-01 specifications; confirm it is what your interface engine expects.

| Situation                                    | PID-3             | PV1                                                                                                    |
|----------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------|
| patient associated                           | id^^^authority^MR | PV1-2 class, PV1-3 unit^room^bed^facility, PV1-19 visit, from the ADT census when the patient is in it |
| no patient, device has a registered location | Unknown^^^U       | PV1-2 U, PV1-3 the device's registered location                                                        |
| no patient, no location                      | Unknown^^^U       | none                                                                                                   |

PID-5 is always ^^^^U (name type unspecified). The gateway does not store or send patient names.

How the gateway decides the patient, per device, from the sources its registration allows:

- **device:** the device reported an identifier (a wristband scanned at the pump);
- **manual:** a person associated them on the Patients page;

- **adt-location:** the ADT census has exactly one patient in the device's bed.

All allowed sources that have an answer must agree, or PID-3 is `Unknown` and the conflict is shown on the Patients page. If ADT location is the only source and would change the patient while the device is running, PID-3 is `Unknown` until a person confirms. The *patient-associated*, *patient-cleared*, *association-conflict* and *association-review* events on the activity page record every decision.

## Rules the receiver can rely on

- **The patient is never guessed.** Unknown is sent as Unknown.
- **Dose rate travels with its basis.** No concentration (or no weight for a weight-based dose) means no dose rate is sent.
- **Units are MDC codes with UCUM alternates** in OBX-6.
- **Timestamps carry a UTC offset.**
- **Codes are labelled.** Every MDC code used is listed in `codes.py` with its source and status. None is yet marked verified against the current RTM tables.

## Sample PCD-01 from the simulator

Registered to ICU 101 A with association by ADT location, after a synthetic admit of TEST0001. First eight segments; the rest are as before.

```
MSH|^~\&|OpenBedside|GATEWAY|EHR|HOSP|20260926142209+0000||ORU^R01^ORU_R01|PD2026092614220900001|P|2.6|||NE|AL|||IHE_PCD_ORU_R01^IHE_PCD^1.3.6.1.4.1.19376.1
PID|||TEST0001^^^HOSP^MR||^U
PV1||I|ICU^101^A|||V0001
OBR|1||PD2026092614220900001^OpenBedside|69985^MDC_DEV_PUMP_INFUS_MDS^MDC|||20260926142209+0000
OBX|1||69985^MDC_DEV_PUMP_INFUS_MDS^MDC|1.0.0.0|||X|||20260926142209+0000|||SIM-0001^OpenBedside-Sim
OBX|2||69986^MDC_DEV_PUMP_INFUS_VMD^MDC|1.1.0.0|||X|||20260926142209+0000|||SIM-0001-A^OpenBedside-Sim
OBX|3||126978^MDC_DEV_PUMP_INFUS_CHAN_DELIVERY^MDC|1.1.1.0|||X|||20260926142209+0000|||
OBX|4||CWE|184508^MDC_PUMP_STAT^MDC|1.1.1.1|^pump-status-infusing^MDC|||R|||20260926142209+0000|||
...
```

## Sample PCD-03 accepted by the simulator

```
MSH|^~\&|TestEHR|TESTFAC|OpenBedside|GATEWAY|20260926142209+0000||RGV^015^RGV_015|T02026092614220900001|P|2.6|||AL|AL|||IHE_PCD_RGV_015^IHE_PCD^1.3.6.1.4.1.1
PID|||TEST0001^^^HOSP^MR
ORC|RE|ORD0001^TestEHR
RXG|1||^heparin|100||263762^MDC_DIM_MILLI_L^MDC|||10|265266^MDC_DIM_MILLI_L_PER_HR^MDC
RXR|IV
OBX|1||69986^MDC_DEV_PUMP_INFUS_VMD^MDC|1.1.0.0|||F|||SIM-0001-A
```

The target module is identified by OBX-18 on the OBX whose OBX-3 is `69986` (VMD). The rate is read from RXG-15 with units in RXG-16 (mL/h only in this release), the volume to be infused from RXG-5.

Reply codes in `RRG^016` MSA-1: `AA` accepted by the simulator; `AE` could not be done (unknown module, module infusing, parameter missing); `AR` refused (not a PIV message, wrong version, or the target is not the simulator).

## ADT the gateway accepts

Identifiers and location only are kept: PID-3 (with PID-3.4 as the assigning authority), PV1-2, PV1-3 and PV1-19. Names and demographics are ignored and never stored.

| Event                                                       | Effect on the census                                  |
|-------------------------------------------------------------|-------------------------------------------------------|
| A01 admit, A04 register, A02 transfer, A13 cancel discharge | patient is at PV1-3                                   |
| A08 update                                                  | PV1-3 if present, otherwise the location already held |
| A03 discharge, A11 cancel admit                             | patient leaves the census                             |
| A12 cancel transfer                                         | patient returns to PV1-6 (prior location)             |
| A40 merge                                                   | MRG-1 becomes PID-3; manual associations follow       |

Every ADT gets an `ACK` with AA, including events the gateway does not use, so the interface engine does not hold them. A message without PID-3 gets AE.

Sample, as sent by `openbedside send-adt --mrn TEST0001 --unit ICU --room 101 --bed A --visit V0001`:

```
MSH|^~\&|TestADT|TESTFAC|OpenBedside|GATEWAY|20260926174324+0000||ADT^A01^ADT_A01|AD2026092617432400004|P|2.6
EVN|A01|20260926174324+0000
PID|1||TEST0001^^^HOSP^MR||TEST^PATIENT
PV1|1|I|ICU^101^A|||||||||||||V0001
```